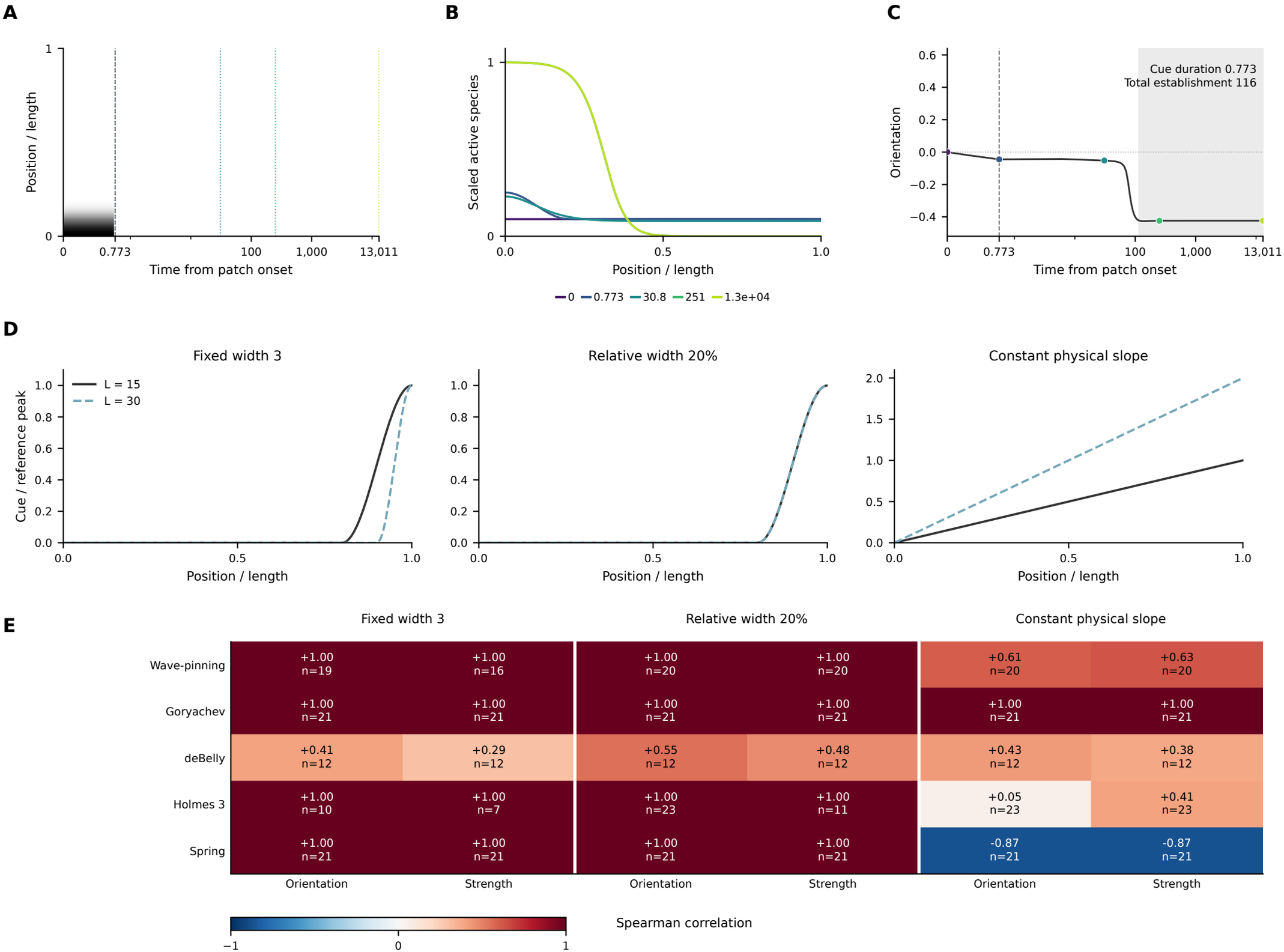

A
Position / length
Time from patch onset
0
0.773
100
1,000
13,011
B
Scaled active species
Position / length
0.0
0.5
1.0
0
0.773
30.8
251
1.3e+04
C
Orientation
Cue duration 0.773
Total establishment 116
Time from patch onset
D
Fixed width 3
Relative width 20%
Constant physical slope
Cue / reference peak
L = 15
L = 30
Position / length
E
Fixed width 3
Relative width 20%
Constant physical slope
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
+1.00 n=19
+1.00 n=16
+1.00 n=20
+1.00 n=20
+0.61 n=20
+0.63 n=20
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+0.41 n=12
+0.29 n=12
+0.55 n=12
+0.48 n=12
+0.43 n=12
+0.38 n=12
+1.00 n=10
+1.00 n=7
+1.00 n=23
+1.00 n=11
+0.05 n=23
+0.41 n=23
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
-0.87 n=21
-0.87 n=21
Orientation
Strength
Orientation
Strength
Orientation
Strength
Spearman correlation
−1
0
1

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

A, Recorded Wave-pinning preparation cue at length 15: a finite left patch, withdrawn at 0.773 native time units. White to black is zero to peak input; coloured vertical lines mark the profile samples in B.

B, Native active profiles from the same preparation, scaled using the common minimum and span of the displayed snapshots. Colours identify recorded times.

C, Orientation from the same preparation. Grey marks basal verification after establishment arrival at about 116; the cue has already ended. The matched time scale in A and C separates input duration from autonomous development.

D, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

E, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.